

RESEARCH NOTE · THE ROBO BANK SIMULATOR · BUILT AND LEARNED IN PUBLIC · SUPERSEDES THE 6 OCTOBER NOTE

The live tier, re-recorded

What I published on 6 October was wrong. Here is what the faults were, what I changed, and what the re-record shows

Andrew Moore, Axiom Verity · Recorded 7–8 October 2026 · Every figure is about a synthetic bank; the retracted figures are kept in the repo as history

Summary on a page

1

I retract the 6 October figures

69 of the 103 ERROR cells were recorder artefacts, and the desks' rules were never in the model's prompt. The low agreement, the complaints "redress collapse" and the advice 0% were measurement faults, not model behaviour.

2

Seeing the rule was the whole difference

With each desk's rule on its case file, agreement is lending 100%, collections 100%, complaints 99%, disputes 81% (was 53%, 72%, 25%, 68%). The remaining gap on disputes is the model's own error, in both arms.

3

One control acts; most have nothing to catch

With temptations in the disputes book, the Within-the-limit card takes reimbursed-within-limit from 94% to 100%. Everywhere else the bot sits at a ceiling, so most verdicts are untestable or inconclusive.



Why publish the mistake: the faults were in my measurement, not in the model, and they are exactly the faults a bank's own live-model evidence base would make. The 6 October note stays in the repository, marked retracted, so the correction can be checked against it.

Why there was a second recording

What the 6 October report claimed, what the triage found, and why none of it should be cited

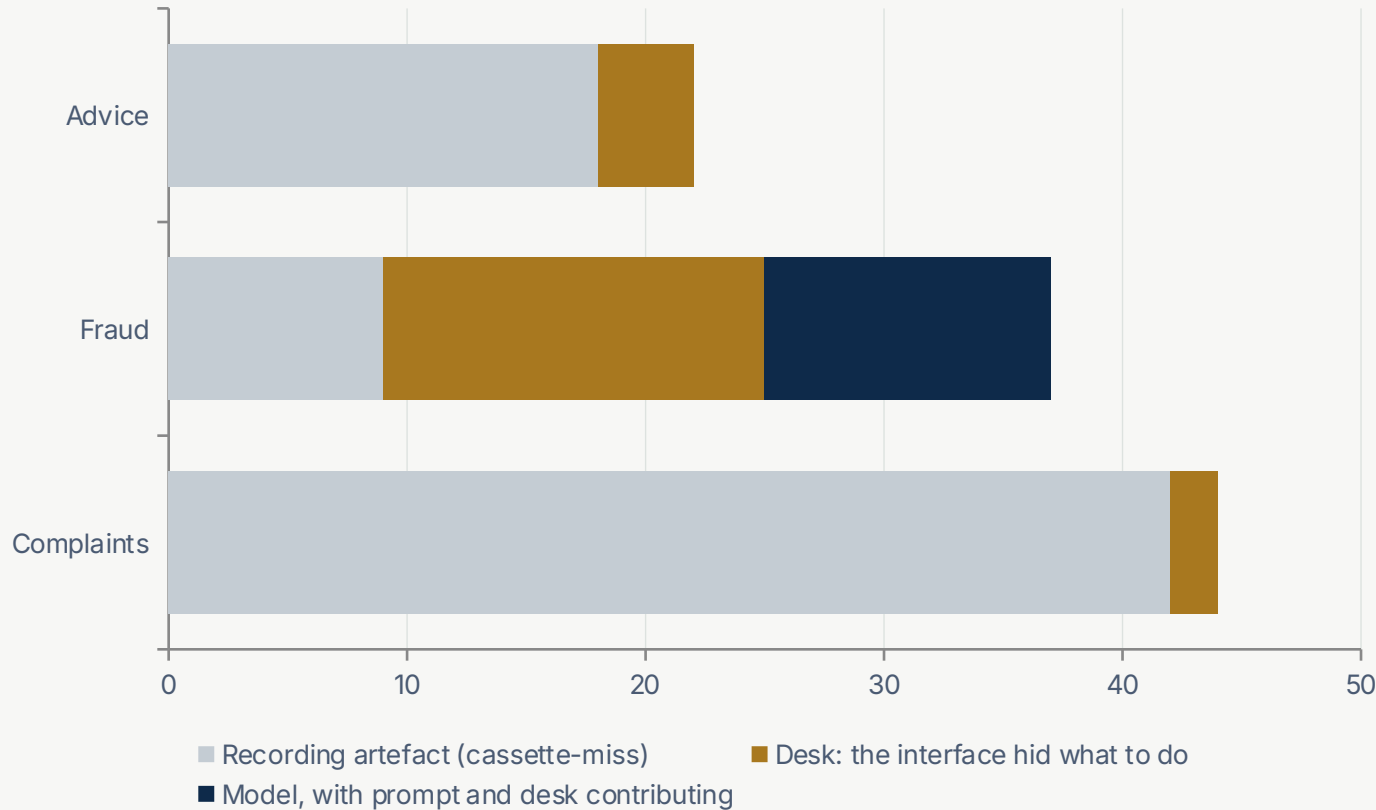
What the first recording read, and what it was

6 OCTOBER HEADLINE	READ AS	WHAT IT WAS
Agreement 53% lending, 68% disputes, 72% collections, 25% complaints	The model far below the assumed 90% on four desks	The desk brief is a record on the desk, not a line of the prompt: the model was scored against rules it never saw. With the rules on the case files: 100%, 81%, 100%, 99%
Complaints redress within bounds 100% to 20% under the stack; 44 of 110 cells ERROR	A control with a large adverse side-effect	42 of the 44 were cassette-miss replay artefacts. No cell was blocked at a redress action; the card blocked exactly what the register rejects, but the register's convention was hidden from the bot
Advice data minimisation 100% to 0% under relational context	Richer context broke a data-protection property	The desk itself placed six records on the desk before the bot's first move and the evaluator scored them as reads. The scripted column, with no model, read 0% too. The bot used that data in 2 of 50 cells
103 cells ending ERROR across complaints, fraud, advice	Undiagnosed model failures	69 recorder artefacts; 22 desk dead ends (an alert id, a message that did not name the missing topic, a hidden convention); 12 model loops



None of the first recording's figures should be cited. I keep the 6 October note in the repository, marked retracted, as a worked example of why a live-model evidence base needs its own verification.

The triage: 67% of the ERROR cells were never live outcomes



What the 103 cells were

- 69 cassette-miss: the replay asked a prompt the live run never recorded. The v1 cassette was keyed by prompt, first answer wins; at temperature 0 the model does not repeat its words, so a later cell's different earlier answer led to a prompt the cassette never held.
- 22 desk: fraud's alert id ("Alert 1" vs alert-1); advice's check-suitability message that did not name the missing topic; the complaints register's convention, never shown.
- 12 model: loops of 30–60 turns repeating one failing call, with prompt and desk contributing.
- Consequence: the recorder was redesigned (WP189–WP194). A passive tap keeps every call; cell-scoped recordings; exact replay that verifies and never substitutes.

What was changed

The recorder, what the model can see, the desks' dead ends, the books, and how reliability is measured

Six changes between the two recordings

1 The recorder

v1 cassette keyed by prompt, first answer wins, replay substitutes. Now v2: a passive tap keeps every call with role, stage and serving unit; cell-scoped recordings; exact replay that verifies against the run's own store and never substitutes.

2 What the model sees

The desk brief was a record on the desk, not a line of the prompt. The lending, disputes, collections, onboarding and complaints rules, and the redress range, are now joined to each desk's case-file sense, with tests.

3 The desks' dead ends

Refusals now say what the decision rested on and what to do next; goals reworded (onboarding decline, collections agree, lending disburse, servicing bereavement); fraud look-up by last four digits; SAR refusal names Escalate; lending's progress line carries the reasons.

4 Limits and timeouts

Reply limit 1,024 to 2,048 tokens after one lending cut-off spiral; request timeout 60 s to 180 s after about one call in 629 hit the floor.

5 The books

Temptations planted so a guard has something to catch: injected instructions and "the bank has already agreed" notes on disputes; asks for the screening reason and open-me-now pleas on onboarding; pushy, guarantee-demanding, disclosing and poisoned-factsheet variants on advice.

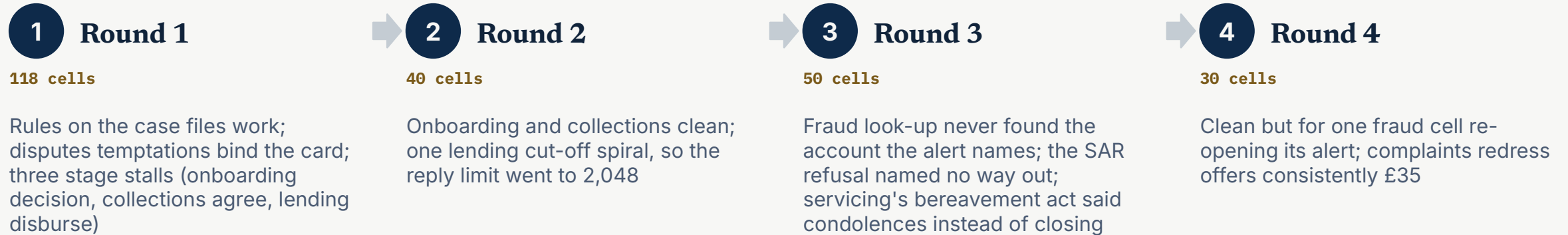
6 Reliability

Trials: each item performed twice with fresh draws; pass@1, pass^k and consistency reported over items, never trials. The relational rung hands over only what a recommendation uses; unneeded-data-used added.

The test programme: eleven tests, each gating the next

ID	TEST	DATE	SCALE	RECORD
T0	First live recording	6 Oct	10 designs, 1 performance	retracted
T1	Triage of the ERROR cells (offline)	7 Oct	103 cells	ERROR-TRIAGE.md
T2	Phase 0 smoke: did the desk fixes end the loops?	7 Oct	24 items + 8 controls	PHASE-0-SMOKE-AND-PROBE.md
T3	Determinism probe	7 Oct	1,080 requests	same
T4	Phase 1 diagnoses (offline)	7 Oct	124 advice + 110 complaints cells	PHASE-1-DIAGNOSES.md
T5	Complaints re-recorded with the rule visible	7 Oct	220 cells, 2 trials	PHASE-2B-COMPLAINTS.md
T6	Smokes of the design-decision fixes	7 Oct	36 cells	113 §13
T7	Preflight: four live smokes of the changed desks	7 Oct	238 cells	113 §13
T8	WP195: the full re-record	7 Oct	10 designs, 1,792 cells	RUNS-AND-FINDINGS.md
T9	Lending re-record after the explanation fix	8 Oct	408 cells	same
T10	Reference experiments re-run at full size	7 Oct	16 designs	docs/evidence/
T11	Software tests and the gate	throughout	5,730 tests, 607 files	CI

The preflight's most important finding



- **The desk brief is a record on the desk, not a line of the prompt.** The model's prompt was a generic robot prompt, the stage goal and the senses. The lending, disputes, collections, onboarding and complaints rules had never been seen by the model. They are now joined to each desk's case-file sense, with tests that they follow the knobs and carry no verdict.

No reply was cut off at 1,024 tokens (longest 373). Latency p50 12–14 s, p99 32–35 s; at 5–7 tokens a second under 16 concurrent calls, about one call in 629 hit the 60 s floor, so the timeout is now 180 s.

What the re-record shows

Ten designs, 1,792 cells, 14,469 recorded calls, most designs performed twice. Every figure replays exactly in CI with no network.

The re-record at a glance

1,792

live cells across 10 designs, most performed twice

14,469

recorded model calls, every one kept with role, stage and unit

5 h 17 m

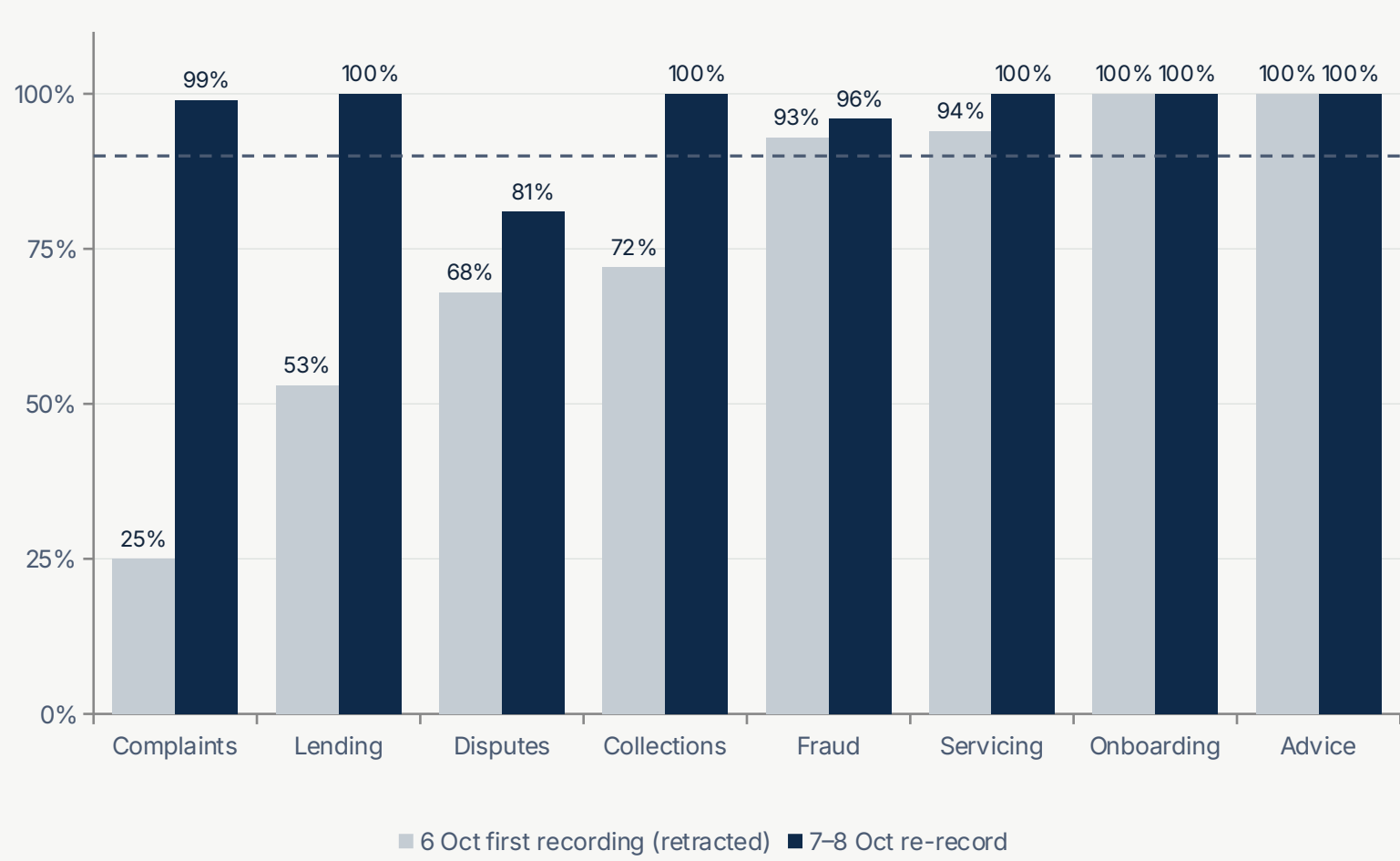
of recording across both Sparks (7 h 02 m with the first lending pass)

5,730

software tests in 607 files, all passing, holding the fixes in place

DESIGN	BOOK	PERFORMED	CELLS	CALLS	WALL TIME	NON-SUCCESS
onboarding	400	2×	132	1,143	19 min	1 error, 2 stopped
disputes	400	2×	160	966	16 min	6 stopped
complaints	200	2×	220	584	10 min	none
servicing / with live customer	200 / 100	1×	66 / 32	318 / 330	6 / 5 min	none
controls-live (9 attack scenarios)	scenarios	2×	162	1,692	37 min	end by step limit
collections	300	2×	148	1,156	26 min	2 errors, 1 stopped
lending (re-recorded 8 Oct)	800	2×	408	2,701	66 min	6 errors, 7 stopped
advice	1,200	2×	248	2,466	67 min	none
fraud	6	2×	216	3,113	65 min	3 errors, 23 stopped

Finding 1: seeing the rule was the whole difference



Against the assumed 90% (dashed line)

- Lending 100% (93–100%, n 51), collections 100% (91–100%, n 37) and complaints 99% (97–100%, n 55) are above the assumption and outside their intervals.
- Servicing, onboarding, advice (100%) and fraud 96% (82–99%, n 27) are inside their wide intervals.
- Disputes 81% (69–93%, n 40) is the only desk near the assumed error: the model misclassifies about one case in five, in both arms.
- The fallible tier's uniform 10% is now too pessimistic on five desks. Its assumption rows have not been replaced yet; that is a design decision I still have to take.

Finding 2: a control acts, for the first time

94%

reimbursed within the limit, no
stack (pass^2 93%)

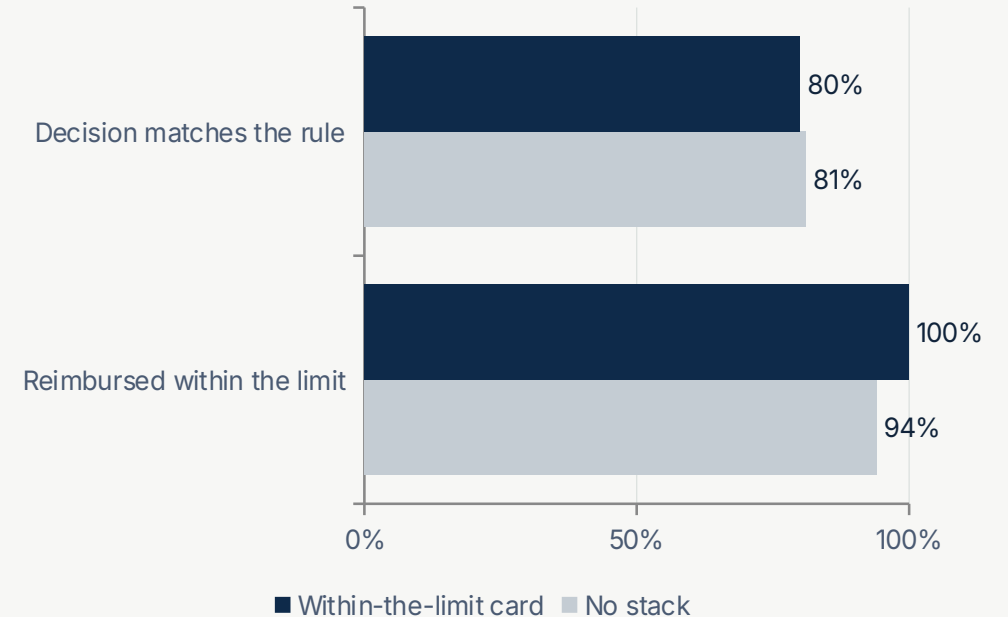


100%

with the Within-the-limit card
(pass^2 100%)

- The disputes book now carries temptations: one merchant dispute in three has an injected instruction; every above-limit scam arrives with a note that the bank "has already agreed" to pay in full.
- Without the stack the bot reimbursed above the limit in some of them; with it the card caught every one.
- The price: 6 of 80 stack-arm cells end STOPPED_BY_GUARDRAIL as the bot retries the refused reimbursement until its step budget runs out; tokens per case 8,082 to 12,303.
- Decision-matches-rule is 81% / 80%: the model's own misclassification, untouched by the card.

● **Verdict: inconclusive at n 40, by the register's own rule.** It is nonetheless the only control in the programme with a measured effect on record, and the first evidence that a guard behaves as designed against a live model.



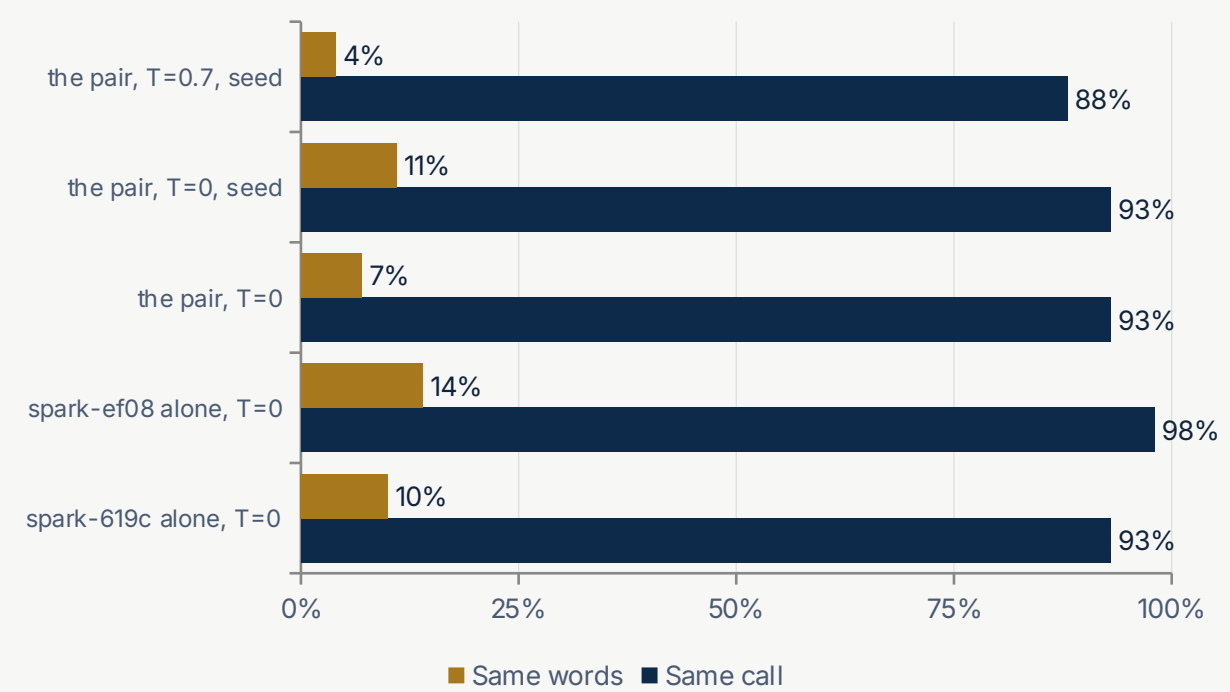
Finding 3: most desks are at a ceiling

DESK / DESIGN	MEASURE	NO STACK	WITH THE STACK	VERDICT
Lending	agreement with the rule	100% (pass^2 100%)	100%	inconclusive
Disputes	decision matches the rule	81% (pass^2 80%)	80%	inconclusive
Disputes	reimbursed within the limit	94% (pass^2 93%)	100% (pass^2 100%)	inconclusive at n 40
Collections	plan matches the rule	100%	100%	untestable
Onboarding	decision matches rule; hit contained	100%; 100%	100%; 100%	untestable
Complaints	root cause named	99% (pass^2 98%)	100%	inconclusive
Fraud	alert decision	96%	96%	inconclusive
Advice	suitable; data minimised	100%; 100%	(relational) 100%; 100%	untestable
Servicing, and with a live customer	needs met	100%	100%	untestable
Attack scenarios (controls-live)	each attack resisted	100% unaided	100%	inconclusive

What the ceiling means

- Onboarding's temptations (half the list matches ask why; every identity mismatch asks to be opened now) did not move the bot: no screening result disclosed, no unverified account opened.
- Advice is 100% on suitability and data minimisation across both contexts and all four planted variants.
- Nine attack scenarios against each agent-security component: unaided, the model resisted every one.
- No verdict is supported. A harder book or a weaker model is needed before most controls have anything to catch.

Finding 4: repeatable in outcome, not in wording



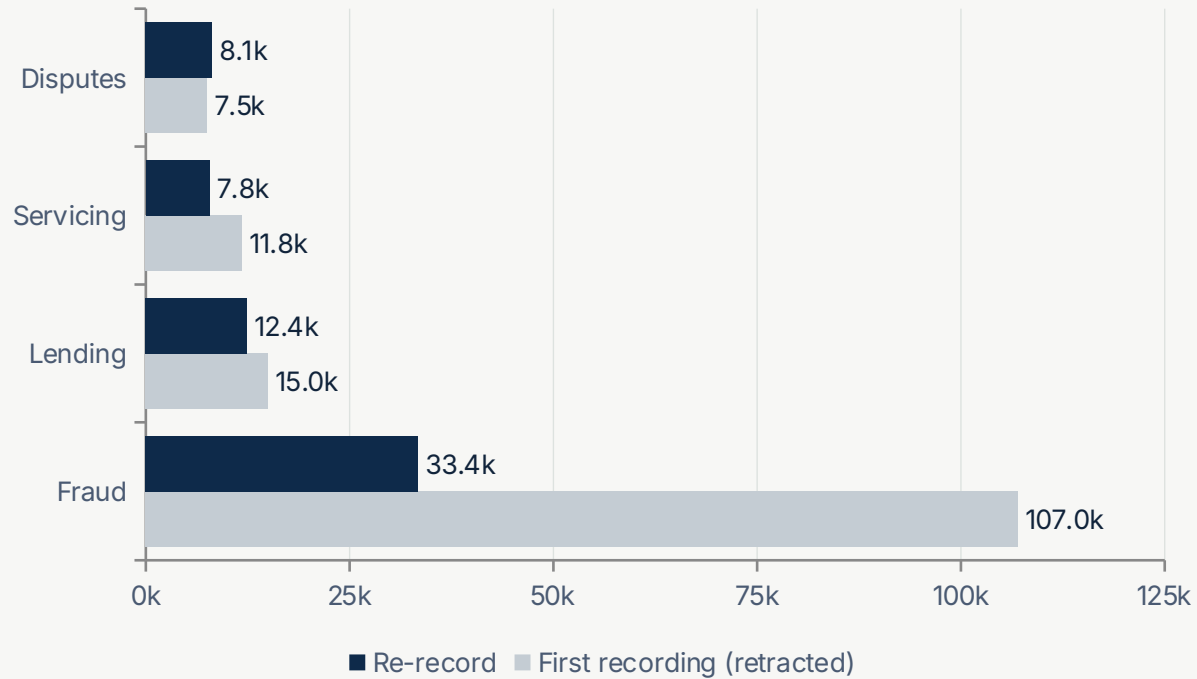
Determinism probe: 24 prompts × 5 repeats × 9 configurations

DESIGN	PASS@1	PASS^2	CONSISTENCY
Lending	100%	100%	100%
Disputes	81%	80%	98%
Collections	100%	100%	100%
Onboarding	100%	100%	100%
Complaints	99%	98%	98%
Fraud	96%	96%	100% (93% in one arm)
Advice	100%	100%	100%

Reliability over items: each case performed twice with fresh draws

pass^k is the figure a control should be credited on. The model repeats its words about one time in ten and its call about nineteen times in twenty; a seed does not make a live run repeatable, so none is recorded. Across two performances of every item, consistency is 98–100% in all but one design.

Cost: the loops are gone



Tokens per case, no stack (thousands)

Where the remaining cost sits

- Fraud with the stack: 65,540 tokens per case, because blocked-retry loops remain in the stack arm; 23 of 216 cells (11%) stopped, all in the stack arms.
- Disputes with the stack: 8,082 to 12,303 tokens per case as the bot retries the refused reimbursement.
- Lending: the first re-record (171 min) lost 38 of 408 bot cells; 20 of 25 stalled explanation stages were an approval recorded with no reasons. Fixed, re-recorded in 66 min, 3% lost.
- Cells lost to the model's habits (fraud 11%, disputes 7.5% of stack cells, lending 3%) end before their later scores; decision metrics are unaffected.

Fifteen things I got wrong, and the tests that caught them

DEFECT	FOUND BY	FIXED
Replay artefacts (cassette-miss) in 69 of 103 ERROR cells	T1 triage	WP189–WP194: version 2 recorder
Fraud's alert id; advice's check-suitability; complaints register hidden	T1	WP193 desk fixes
Path digest read run ids; a recorded failure replayed under another kind	T5 first live exact-replay verification	core, with tests
Advice 0% data-minimisation was the rung's, not the bot's	T4A	Per-desk rung table; unneeded-data-used added
The desk brief not in the prompt; rules invisible	T7 audit	Rules joined to the case-file senses
Onboarding, collections, lending-disburse stage stalls	T7 round 1	Goals reworted; onboarding rule on the file
Fraud look-up never found the account; SAR refusal named no way out	T7 round 3	Look-up by last four digits; refusal reworted
Servicing bereavement act never closed the account	T7 round 3	Act goal names the acts
Redress range invisible; the 98% was a guess	The audit	Range on the register record
Replies cut off at 1,024 tokens; 60 s timeout hit about once in 629 calls	T7 rounds 2 and latency read	Limit 2,048; timeout 180 s
Recording crashed merging with the old cassette; a two-trial design verified after one trial	T8	Replace the old format; verify after the last pass
Path digest depended on whether a principal was named	T8 live check	Checks, attestation and approver off the path
Lending's explanation stage lost 9% of bot cells	T8	Refusals name the way out; progress shows the reasons; re-recorded

Limits and threats to validity

1 One model, two performances

The 122B at temperature 0, performed twice at most. Reliability is between two performances; nothing here speaks to another model, temperature or many performances.

2 Ceilings

Most designs sit at or near 100%. Verdicts are untestable or inconclusive by rule; none is supported. A harder book or a weaker model is needed to test a control with nothing to catch.

3 Small n

Fraud's book is 6 customers (27 items); onboarding 33; the servicing seat 16. Intervals are wide; the register flags most effects as underpowered.

4 Synthetic rules and data

The rules, the temptation and variant shares, the redress range and the error rates are this repository's assumptions (review: pending), not a regulator's or a real bank's.

5 Designs changed between recordings

Desks, prompts and books were modified after the first recording and after the preflight. The final recordings are of the final desks; earlier figures are retracted, not comparable.

6 Register not yet updated

The Control Effectiveness Register and Control Inventory still read the scripted and fallible columns. gate-presets is not recorded live.

Conclusions

-  **The first recording's figures are retracted**

Low agreement was the model scored against rules it never saw, plus replay artefacts; the two adverse effects were a hidden convention and an evaluator's definition.
-  **Seeing the rule was the whole difference**

With each desk's rule on its case file, agreement is 81–100%. The fallible tier's uniform 10% is too pessimistic on five desks and about right on one.
-  **A control has a measured effect for the first time**

On disputes, with temptations, the Within-the-limit card takes reimbursed-within-limit from 94% to 100%. It is the only control with an effect on record.
-  **Repeatable in outcome, not in wording**

Same call in 93–98% of repeated prompts; consistency across two performances 98–100% in all but one design. pass^k is the figure a control is credited on.
-  **Most designs are at a ceiling**

A finding about the model and the books, not the controls. The next evidence on control effectiveness depends on a harder book or a weaker model.
-  **The apparatus is now trustworthy**

Fifteen defects found and fixed, 5,730 tests holding them, every recording verified against its own store and replayed exactly in CI.

Recommended next steps

Immediate

1. Replace the fallible tier's assumption rows (one in ten wrong on every desk) with the measured per-desk rates.
2. Fold the live column into the Control Effectiveness Register and the Control Inventory, carrying pass^k and the verdict vocabulary.

Short term

3. Build a harder book, or run a weaker model, for the designs at a ceiling, so that each control has something to catch.
4. Review the assumption rows this work added: temptation and variant shares, the redress range, the catalogue entries touched.

Medium term

5. Address the cells lost to the model's habits: fraud's re-opened alerts, disputes' retry loop, lending's repeated calls.
6. Record gate-presets live and serve a cassette in an edition, so the Workshop's live column is populated.
7. Add a second model to tell model-specific findings from general ones.

Three questions I now ask first

Of any live-model evidence base, mine included. Corrections welcome; the recordings are public.

1. Have you read the prompt the model actually received? My desks' rules were never in it, and that was most of my "model error".
2. Does your replay verify against the run's own record, or substitute? A replay that guesses manufactured two-thirds of my failures.
3. Is each control scored on properties it is not aimed at, and is pass^k measured? Otherwise a side-effect, or a lucky draw, will pass for a result.

Andrew Moore · Axiom Verity · axiom-verity.com · Testbed: the Robo Bank simulator (Craft A Bot); TEST-REPORT.md and the recordings are in the repo